

Exercise 3: Google App Engine SDK and Web App Development

Aim

To install the Google App Engine SDK and develop a Hello World / simple web application using Python.

Procedure

Step 1: Install Google Cloud CLI & Python

Ensure Python 3 is installed on the system.

```
ubuntu@ubuntu:~$ sudo apt install python3
python3 is already the newest version (3.14.3-0ubuntu2).
python3 set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 3
```

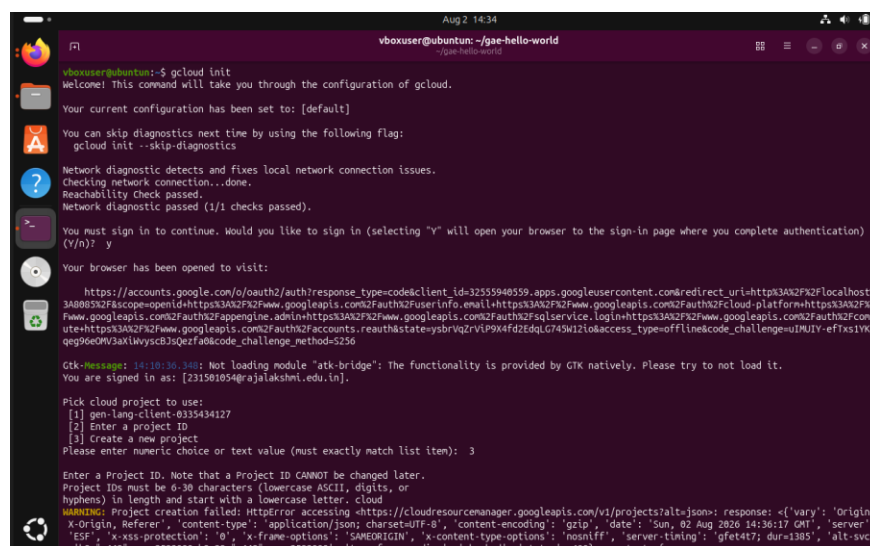
Download and install the Google Cloud CLI (contains the modern App Engine SDK) from cloud.google.com/sdk/docs/install:

```
curl -O https://dl.google.com/dl/cloudsdk/channels/rapid/downloads/google-cloud-cli-  
linux-x86_64.tar.gz  
tar -xf google-cloud-cli-linux-x86_64.tar.gz  
./google-cloud-sdk/install.sh
```

Step 2: Initialize GCloud & Install Components

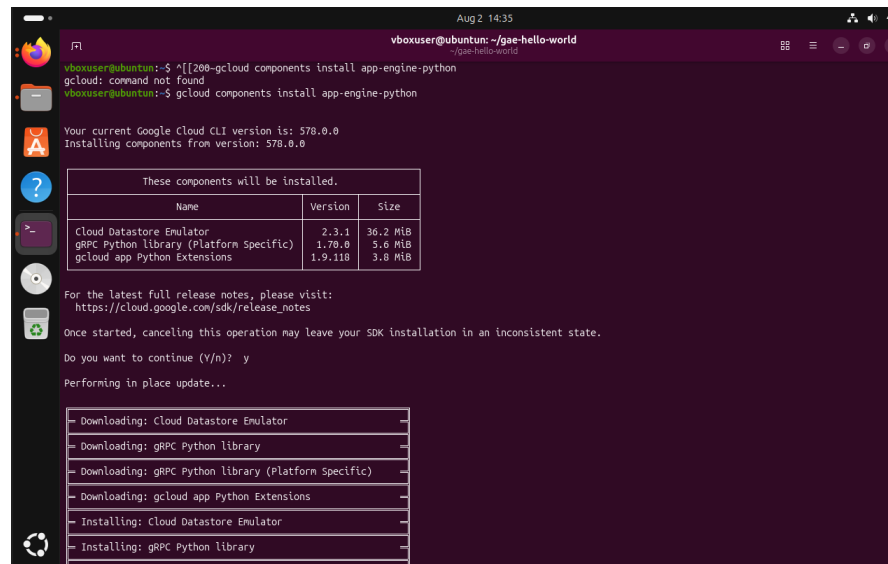
```
gcloud init
```

Sign in to your Google account when prompted.



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gcloud components install app-engine-python



Step 3: Create the App Directory

```
mkdir gae-hello-world
cd gae-hello-world
```

Step 4: Create Application Code (main.py)

```
from flask import Flask
app = Flask(__name__)

@app.route("/")
def hello():
    return "Hello World from Google App Engine!"

if __name__ == "__main__":
    app.run(host="127.0.0.1", port=8080, debug=True)
```

Step 5: Create Configuration File (app.yaml)

```
runtime: python39
handlers:
- url: /.*
  script: auto
```

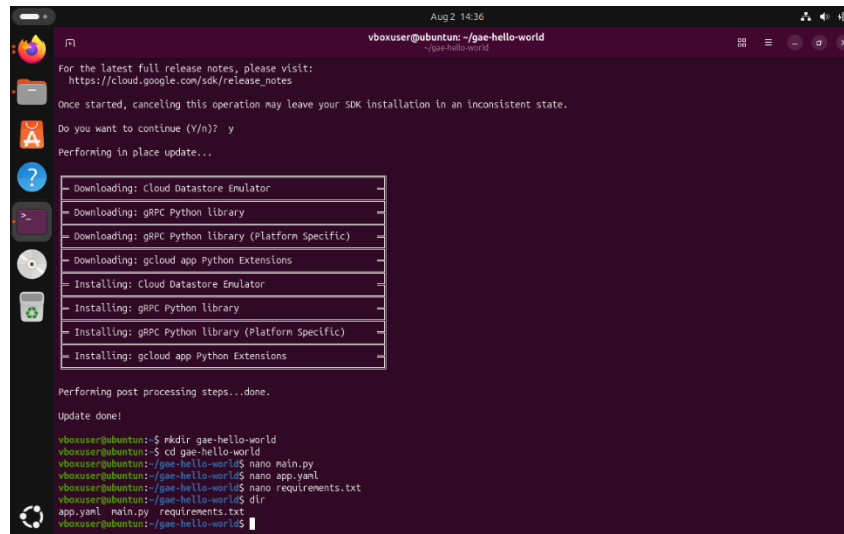
Step 6: Create requirements.txt

```
Flask==2.0.1
```

Output

Files created in project directory (main.py, app.yaml, requirements.txt):

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A terminal window titled 'vboxuser@ubuntu: ~/gae-hello-world' showing the installation of the Google App Engine SDK. The terminal displays a progress bar with the following items: Downloading: Cloud Datastore Emulator, Downloading: gRPC Python library, Downloading: gRPC Python library (Platform Specific), Downloading: gcloud app Python Extensions, Installing: Cloud Datastore Emulator, Installing: gRPC Python library, Installing: gRPC Python library (Platform Specific), and Installing: gcloud app Python Extensions. Below the progress bar, it says 'Performing post processing steps...done.' and 'Update done!'. The terminal also shows the user navigating to the 'gae-hello-world' directory and creating files 'main.py', 'app.yaml', and 'requirements.txt'.

```
vboxuser@ubuntu: ~/gae-hello-world
For the latest full release notes, please visit:
https://cloud.google.com/sdk/release_notes

Once started, cancelling this operation may leave your SDK installation in an inconsistent state.
Do you want to continue (Y/n)? y
Performing in place update...

- Downloading: Cloud Datastore Emulator
- Downloading: gRPC Python library
- Downloading: gRPC Python library (Platform Specific)
- Downloading: gcloud app Python Extensions
- Installing: Cloud Datastore Emulator
- Installing: gRPC Python library
- Installing: gRPC Python library (Platform Specific)
- Installing: gcloud app Python Extensions

Performing post processing steps...done.
Update done!

vboxuser@ubuntu: $ mkdir gae-hello-world
vboxuser@ubuntu: $ cd gae-hello-world
vboxuser@ubuntu: ~/gae-hello-world$ nano main.py
vboxuser@ubuntu: ~/gae-hello-world$ nano app.yaml
vboxuser@ubuntu: ~/gae-hello-world$ nano requirements.txt
vboxuser@ubuntu: ~/gae-hello-world$ dir
app.yaml  main.py  requirements.txt
vboxuser@ubuntu: ~/gae-hello-world$
```

Result

Thus, the Google App Engine SDK was installed and a simple Hello World web application was developed using Python and Flask.